Strüver Model: Universal Stability Kernel (USK)

Axiomatic Compression, Deterministic Operators, Falsifiability, Hardware Mapping

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This document defines a deterministic stability architecture. It does not claim new physics. It defines admissibility conditions for irreversible system actions. Validity requires independent evaluation and external falsification.

Contents

1 Purpose

The Universal Stability Kernel (USK) is a domain-invariant, deterministic admissibility layer.
Its purpose is:

- Normalize physical telemetry into bounded proxies.
- Aggregate proxies into functional axes.
- Compute a stability state $\Phi(t)$.
- Derive a hazard value $h(t)$.
- Gate irreversible system actions via invariant enforcement.

USK is not a subsystem physics model. It is a commit-level execution constraint.

2 Minimal Ontological Standard

Axiom 1 (Observable State)**.** *There exists a measurable state vector $x(t) \in \mathbb{R}^n$, derived from physical sensors.*

Axiom 2 (Evidence Validity)**.** *A measurement is admissible only if:*

$$\textit{freshness} \leq \Delta t^{max} \quad \wedge \quad \textit{conformance} = 1$$

Otherwise it is undefined.

Axiom 3 (Irreversible Commit)**.** *Certain actions cross a commit boundary after which system trajectory cannot be deterministically reverted within the operational horizon.*

3 Proxy Normalization

For each telemetry atom a_i:

$$p_i(t) = \begin{cases} \operatorname{clip}\left(\left(\frac{x_i(t)-\mu_i}{\sigma_i}\right) s_i + b_i, 0, 1\right) & \text{if valid} \\ \perp & \text{otherwise} \end{cases}$$

All proxies are bounded in $[0, 1]$ and fixed-point compatible.

4 Functional Axis Model

Let $F_1, \ldots, F_k$ be functional axes.

$$F_j(t) = \sum_{i \in I_j} w_i p_i(t)$$

with $\sum w_i = 1$.
Global stability:

$$\Phi(t) = \begin{cases} \sum_{j=1}^{k} W_j F_j(t) & \text{if all valid} \\ \perp & \text{otherwise} \end{cases}$$

5 Corridor Operator

Relative axis contribution:

$$\rho_j(t) = \frac{F_j(t)}{\sum_{\ell=1}^{k} F_\ell(t)}$$

Corridor condition:

$$\rho_j^{min} \leq \rho_j(t) \leq \rho_j^{max}$$

Deviation measure:

$$D_{band}(t) = \sum_{j=1}^{k} \max\{0, \rho_j^{min} - \rho_j(t), \rho_j(t) - \rho_j^{max}\}$$

Stability is a band condition, not a point condition.

6 Hazard Operator

Let:

$$O(t) = \max_j \rho_j(t)$$

$$G_\Delta(t) = |\Phi(t) - \Phi(t - \Delta)|$$

Hazard:

$$h(t) = h_{max}\left(1 - \exp\left(-(k_1 D_{band} + k_2 O + k_3 G_\Delta + k_4 D_{in})\right)\right)$$

If evidence is undefined:

$$h(t) = h_{max}$$

No evidence implies maximal risk.

7 Invariant-Bound Commit Gate

Given invariant set $\mathcal{I}$:

$$ALLOW \iff (\forall I_\ell = SAT) \wedge (h(t) \leq \bar{h})$$

$$REFUSE \iff (\exists I_\ell \in \{VIOL, UND\})$$

$$ESCALATE \iff \neg ALLOW \wedge \neg REFUSE$$

Execution is blocked unless evidence sufficiency and invariants are satisfied.

8 Falsifiability Criteria

USK makes risk-bearing predictions:

1. Hazard increases prior to irreversible failure events.
2. Axis re-balancing reduces hazard within defined time bounds.
3. Out-of-sample performance exceeds naive trend baselines.

If these conditions fail under controlled testing, the rule is invalid in that domain.

9 Hardware Mapping

USK is implementable as:

- Fixed-point proxy arithmetic
- Deterministic finite state machine
- Non-bypassable commit gate
- Replay-verifiable evaluation layer

No stochastic dependency is required.

10 Conclusion

The Strüver Universal Stability Kernel defines:

- Deterministic admissibility
- Evidence-bound execution
- Corridor-based stability
- Hazard-derived gating

It is not a belief system. It is a structural execution constraint.
Scientific validity depends solely on independent falsification.